Learning Objectives

- To learn the usage and importance of various types of fuels.
- To learn about various solid, liquid and gaseous fuels.

3.0 INTRODUCTION

For a healthy body, we consume solid food, liquid food and pure air. Similarly, for an engine to operate, it requires fuel.The heat energy released during burning the fuel with air, is converted into mechanical energy via a heat engine. This mechanical energy gives the required tractive force to move the vehicle in the forward direction.



Figure 3.1.1 Solid Fuels

3.1 FOSSIL FUELS

Fossils fuels are available as Solid, Liquid and Gaseous state. Figure 3.1 shows Fossil Fuels.



Figure 3.1 Fossil Fuels

3.1.1 Solid Fuels

Solid fuel refers to various types of solid material that are used as fuel to produce energy and provide heating, usually released through combustion. Solid fuels include wood, coalare mining under the earth. Initially solid fuels are used in steam engines and boilers. They release less heat energy and emit more ash and emissions. However, the solid fuels are not used in modern automobiles. Figure 3.1.1 shows Solid Fuel.

3.1.2 Liquid Fuels

Many liquid fuels play a primary role in transportation. Liquid fuels are easy to store, easy to transport, and can be handled with relative ease. They release more heat energy and less emission. Petrol and Diesel fuels are widely used for automobiles. Figure 3.1.2 shows Liquid Fuels.



Figure 3.1.2 Liquid Fuels

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